MOVEMENT OF PARTICLES 4

When a crystal of copper (II) sulfate is placed in a beaker of water, over a period of time the water will turn blue and the crystal will disappear.

1 Complete the sentence: *The copper sulfate has* _____ *in the water to form a* _____.



Figure 1

2 The crystal (solid) and the water (liquid) are both made up of particles. Use the diagrams in figure 1 to explain why the water turns blue and the crystal disappears.

_____

_____

3 Can you see the individual copper sulfate particles after the crystal has dissolved? Explain your answer.

_____

Diffusion is the movement of particles among one another so that they become evenly mixed.

Bromine (Br) forms a red-brown vapour and is denser than air.

4 What colour is air? _____

When a gas jar of air is placed over a jar of bromine vapour so that the gases can mix together, the different gases look the same after a few minutes. The colour of the gases is paler than the colour of the original bromine gas. This is shown in figure 2.

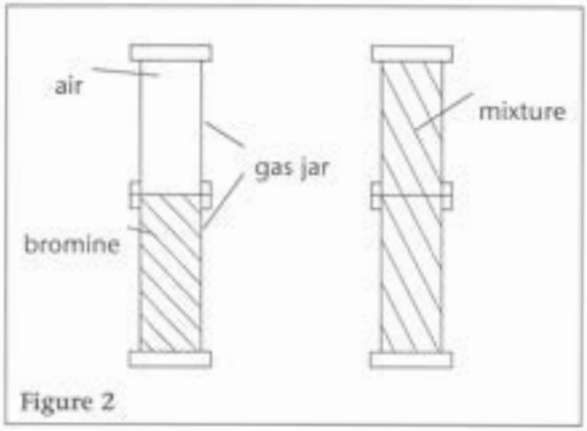


Figure 2

5 Complete the sentences:

Air and bromine vapour are both made up of tiny moving _____. They bump into each other and move in all directions so that they eventually become evenly mixed. This is an example of _____.

6 Why is the mixture of gases paler in colour than the original bromine gas?

_____

7 A teacher placed a few millilitres of perfume on a glass dish at the front of the class. Then she asked the students to raise their hands when they could smell the perfume. Describe what would happen and explain why.

_____

_____

8 Generally speaking, diffusion takes place more quickly with gases than with liquids. Give a reason for this.

9 Kinetic energy is the energy of motion. True or false?

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Brownian motion

Robert Brown was a Scottish botanist on Matthew Flinders' expedition to Australia (1801–05). On this expedition, Brown collected around 1700 new plant species. In 1827, he was examining pollen grains under a microscope. The pollen grains were suspended in water and he observed that the tiny particles were moving in a continuous, erratic manner. His first explanation was that pollen grains, being from a living organism, were moving of their own accord. This theory was disproved when a similar effect resulted from suspending in water small inorganic particles such as toothpaste and graphite (carbon).

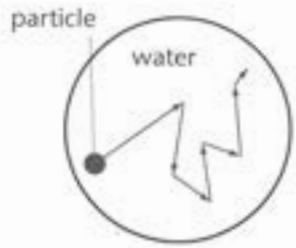


Figure 3

Water molecules (which are too small to be seen) have **kinetic energy** (i.e. they are moving). These molecules hit the suspended particles at irregular intervals and from irregular directions.

The collisions between the water molecules and the particles make the particles move. This erratic motion of the visible particles is sometimes called 'the drunkard's walk' (figure 3).

For this effect to be visible, particle size is very important. The particles have to be small enough to be deflected by water molecules but large enough to be seen under a microscope. It is easier to use smoke trapped in a small glass box and examine it under a microscope.

Light is focused onto smoke by a lens (figure 4). The smoke cell is sealed so that draughts do not affect the smoke. The smoke particles are constantly bombarded by the invisible air molecules, causing a random, 'jittery' motion.

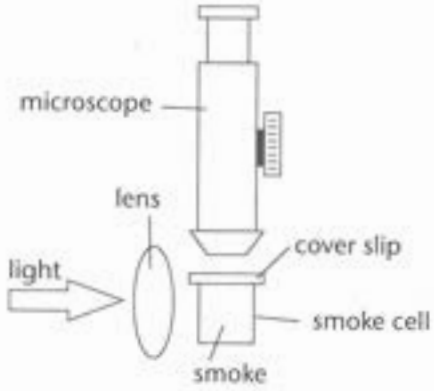


Figure 4

10 For the apparatus shown in figure 4, state the function of the:

a smoke cell and cover slip _____

b lens _____

c microscope _____

d light _____

11 Give a brief definition for each of the following terms.

a botanist _____

b inorganic _____

c fluid _____

d random _____

e kinetic _____

f particle _____

g vapour _____

12 What does the following define? _____

The movement of particles from an area of greater concentration to an area of lesser concentration – spreading out equally to fill all the space available.

13 *Particles have to be small enough to be deflected by water molecules.* Explain this statement in regard to Brownian motion.

14 Give your own definiton for *Brownian motion.*

15 Circle the correct word in brackets to complete the sentence: *Brownian motion provides evidence that the (inorganic, random, kinetic) model is accurate.*

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